Rev	Date	Eng	Changes
AA	2026-11-11	NB	First build (Rev0); derived from Sofar mote 000639-AB

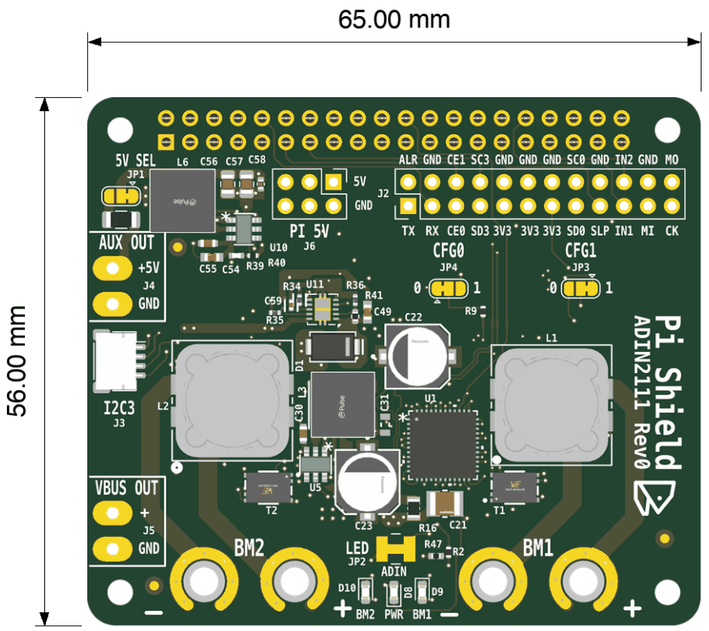
Nereus_Pi_Shield_Adin2111

Variant: Standard

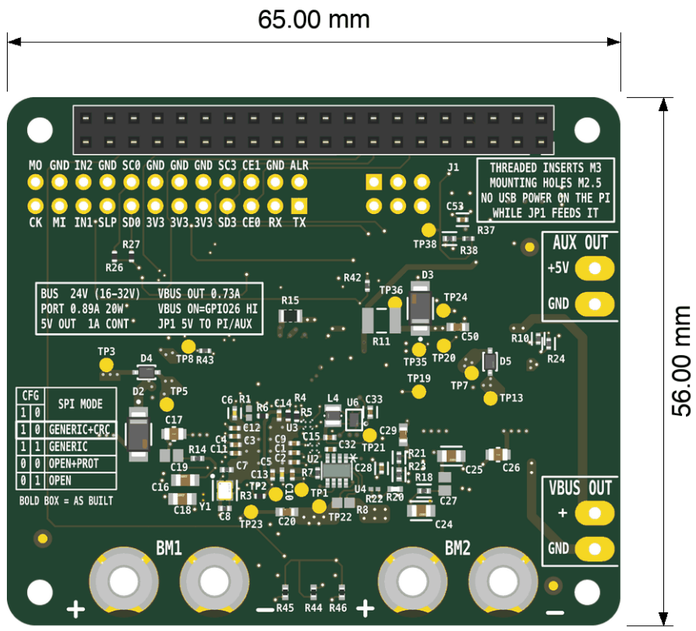
P/N:PCB-000001-AA

Index

1 Cover
2 Top-Level
3 ADIN2111 (Single-Pair Ethernet)
4 Load Switch (payload port)
5 PoDL (Power over Data Line)
6 Bus Power Monitor
7 Power (buck converters)



TOP VIEW



BOTTOM VIEW (seen from below)

Outline 65.00 x 56.00 mm, corners R3.0, 1.6 mm thick, 6 copper layers. 4 mounting holes 2.70 mm on 58.00 x 49.00 mm. Views at 1.54 : 1 on A3.

I2c Address Table

Device	I2C Address
Power Monitor	0x41

DIMENSIONS: 65.00 x 56.00 x 1.6 mm

Top-Level Schematic
File: Nereus_Pi_Shield_Adin2111_TopLevel.kicad_sch

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https://creativecommons.org/licenses/by-nd/4.0/legalcode

Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
Nereus_Pi_Shield_Adin2111 · PCB-000001-AA

Sheet: /
File: Nereus_Pi_Shield_Adin2111.kicad_sch

Title: Cover Page

Size: A3	Date: 2026-11-11	**Rev: AA**
KiCad E.D.A. 10.0.7		Id: 1/7

I2C1 GPIO2/3 = LiFePO4wered only (0x43). Pi's 1.8k pull-ups.

I2C0 GPIO0/1 = shield: INA232 0x41, ADS1115 0x48 (bluewired). R26/R27 4.7k to PI_3V3.
Keep 0x50 free (HAT EEPROM probe at boot).

I2C3 GPIO22/27 = software I2C, Qwiic. Qwiic power and pull-ups on PI_3V3.

J2 breakout 3V3 pins (9, 11) = PI_3V3. Shield 3V3 is for shield parts only.

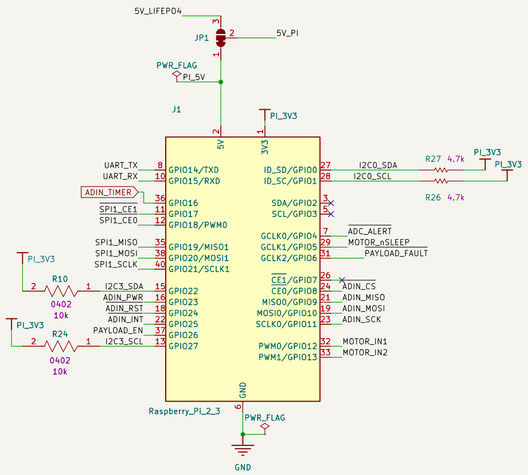
ADS1115: AIN0 = MP6550 VISEN (200 mV/A), AIN1 = shaft pot, AIN2 = spare. AIN3 unused.

JP1 power mode (middle pad = shield 5V. 5V_PI):

Mode 1 (as built): pads 1–2 bridged -> shield powers the Pi through J1 pin 2.

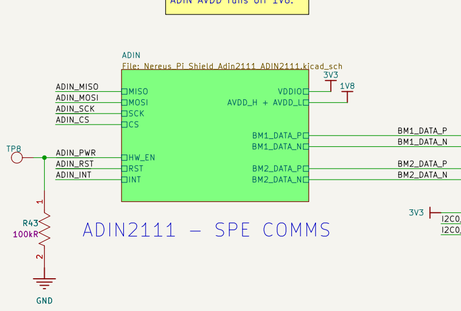
Mode 2 (LiFePO4wered): cut 1–2, bridge 2–3 -> shield 5V goes out on the JST-PH
  to the LiFePO4wered INPUT. Never wire it to the LiFePO4wered 5V output.

Never plug a 5 V source into the Pi's USB while 1–2 is bridged.

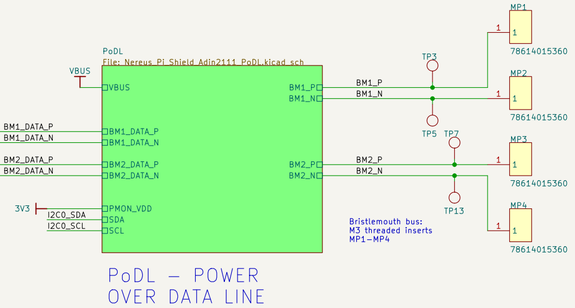


RASPBERRY PI — 40-PIN HEADER

Shield logic runs off 3V3;
ADIN AVDD runs off 1V8.

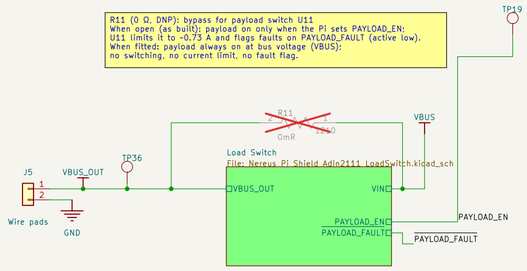


ADIN2111 — SPE COMMS



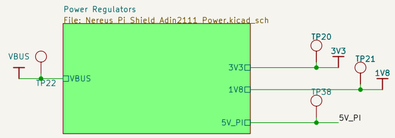
PoDL — POWER
OVER DATA LINE

R11 (0 Ω, DNP): bypass for payload switch U11
When open (as built): payload on only when the Pi sets PAYLOAD_EN;
U11 limits it to ~0.73 A and flags faults on PAYLOAD_FAULT (active low).
When fitted: payload always on at bus voltage (VBUS);
no switching, no current limit, no fault flag.



Payload power out: J5, JST GH 1.25 mm (D23).
Pin 1 VBUS_OUT: 16–32 V bus, switched by U11,
~0.73 A limit. Pin 2 GND.
Mate the cable before potting.

LOAD SWITCH — PAYLOAD POWER



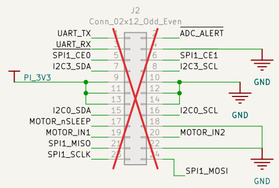
POWER — BUCKS

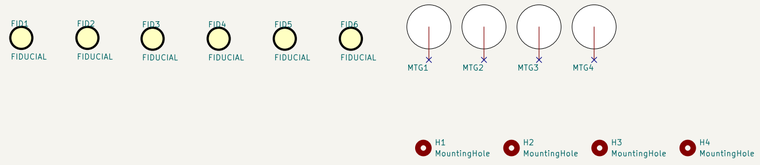


PI 3V3 REGULATED POWER

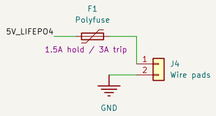


HEADER BREAKOUT FOR DEVELOPMENT

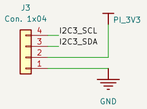




POWER PATH FOR EXTERNAL SHIELD



Qwiic CONNECTOR / I2C3 SOFTWARE EMULATION



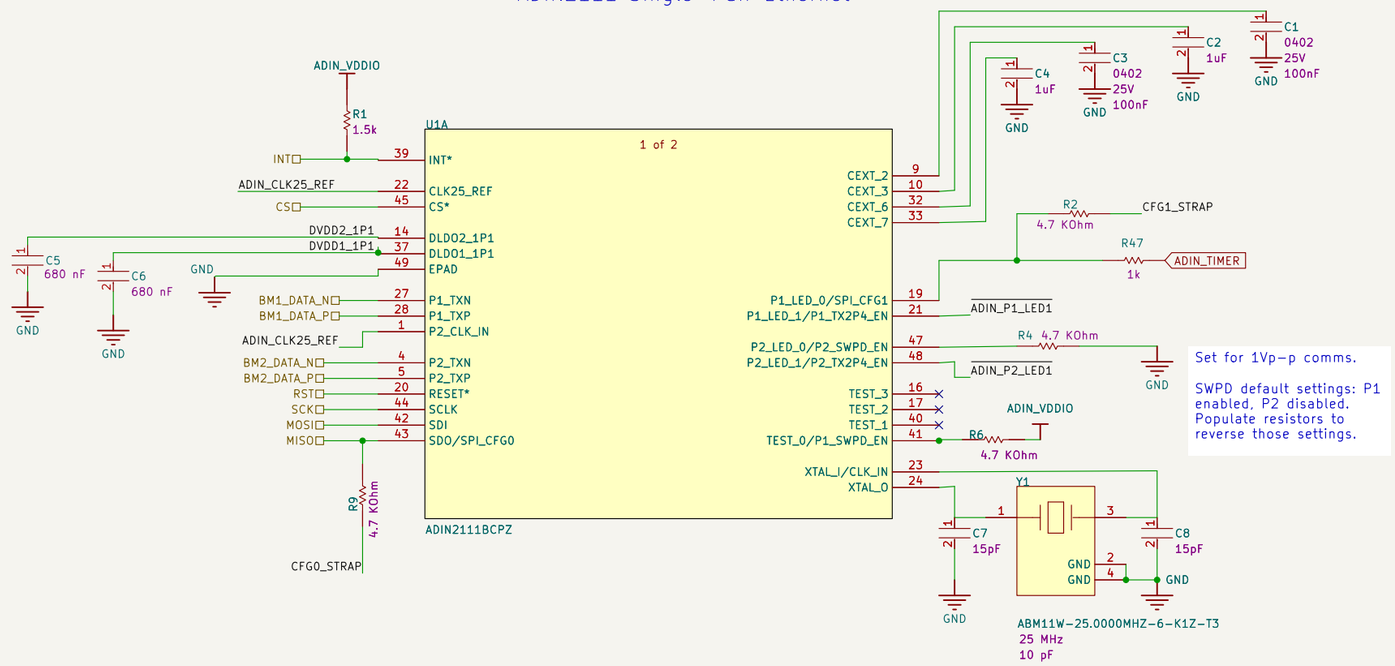
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
Nereus_Pi_Shield_Adin2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/
File: Nereus_Pi_Shield_Adin2111_TopLevel.kicad_sch

Title: Top-Level

Size: A2 | Date: 2026-11-11 | **Rev: AA**
KiCad E.D.A. 10.0.7 | Id: 2/7

ADIN2111 Single-Pair Ethernet



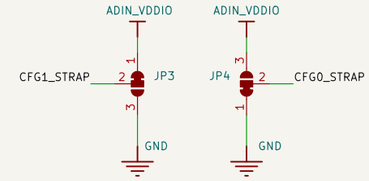
Set for 1Vp-p comms.

SWPD default settings: P1 enabled, P2 disabled. Populate resistors to reverse those settings.

ADIN2111 SPI CFG Straps

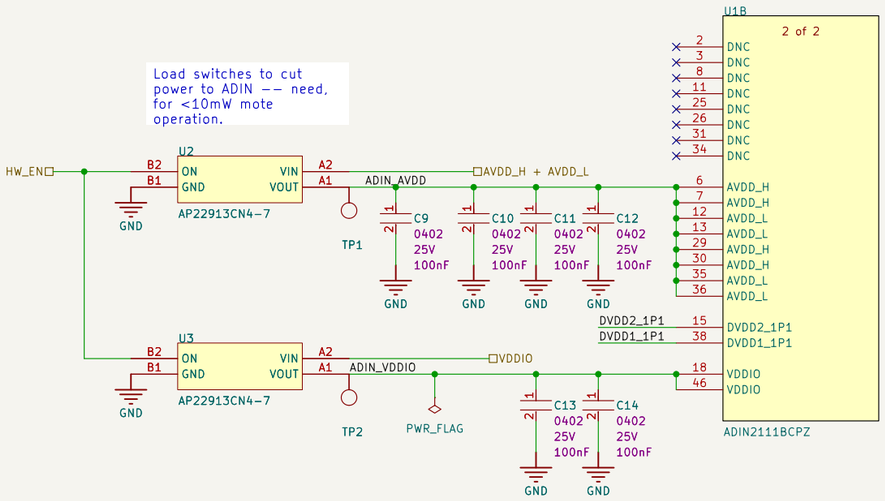
SPI CFG Mode	JP3 (SPI_CFG1, pin 19)	JP4 (SPI_CFG0, pin 43)
Generic SPI + CRC (default)	1-2 (VDDIO)	1-2 (GND)
Generic SPI, no CRC	1-2 (VDDIO)	2-3 (VDDIO)
OPEN Alliance + protection	2-3 (GND)	1-2 (GND)
OPEN Alliance, no protection	2-3 (GND)	2-3 (VDDIO)

SPI_CFG1 (pin19) defaults "1"
cut trace and solder bridge to get "0"

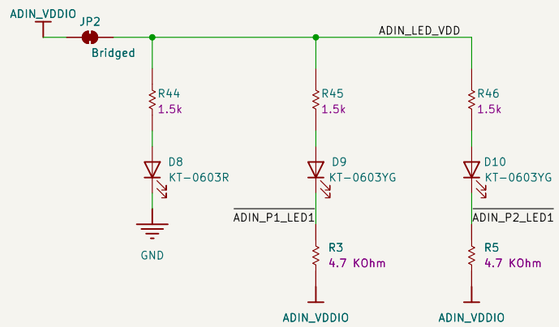


CFG0_STRAP (pin43) defaults "0",
cut trace and solder bridge to get "1"

ADIN2111 supplies: U2 (AVDD) and U3 (VDDIO) switched on by ADIN_PWR



ADIN2111 status LEDs: power, port 1 / port 2 link-activity



ADIN status LEDs (D25): D8 red = ADIN powered (ADIN_VDDIO on).
D9 / D10 yellow-green = port 1 / port 2 link/activity, ADIN pins 21 / 48, active low.
Off until the Pi enables each port's LED_1 over SPI; port 2 is also asleep until woken.
R3 / R5 keep the TX2P4_EN straps high (1.0 V p-p). Cut JP2 to disable all three LEDs.

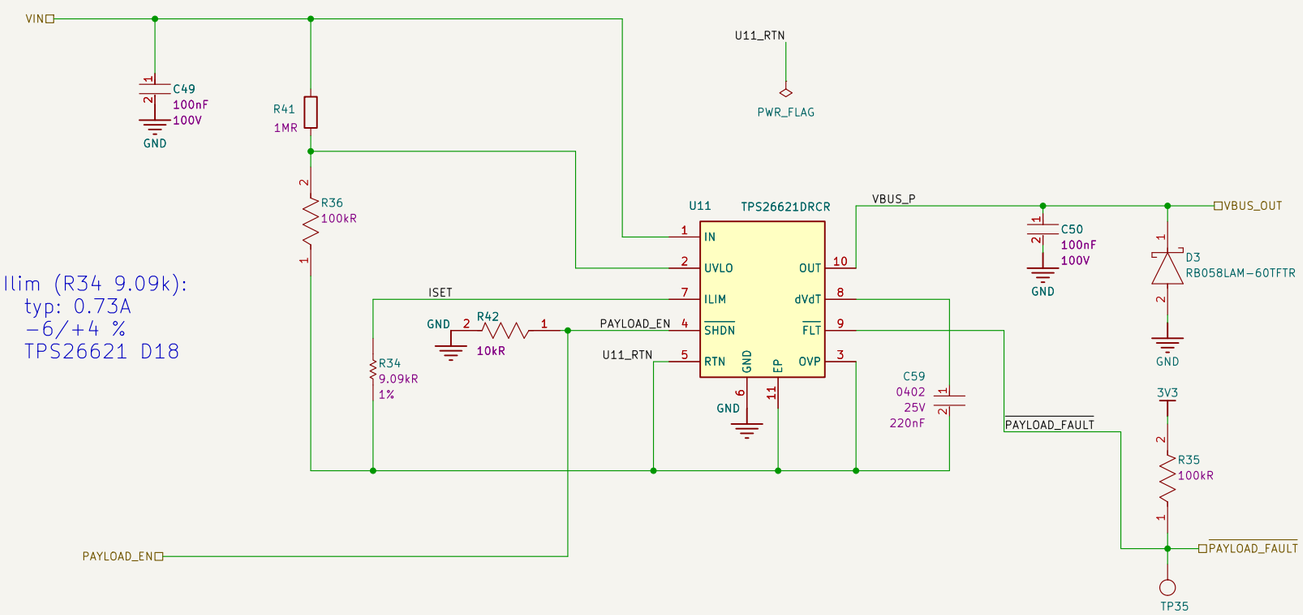
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
Nereus_Pi_Shield_Adin2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/ADIN/
File: Nereus_Pi_Shield_Adin2111_ADIN2111.kicad_sch

Title: ADIN2111

Size: A3	Date: 2026-11-11	**Rev: AA**
KiCad E.D.A. 10.0.7		Id: 3/7

Load Switch for VBUS



Ilim (R34 9.09k):
typ: 0.73A
−6/+4 %
TPS26621 D18

Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
Nereus_Pi_Shield_Adin2111 · PCB-000001-AA

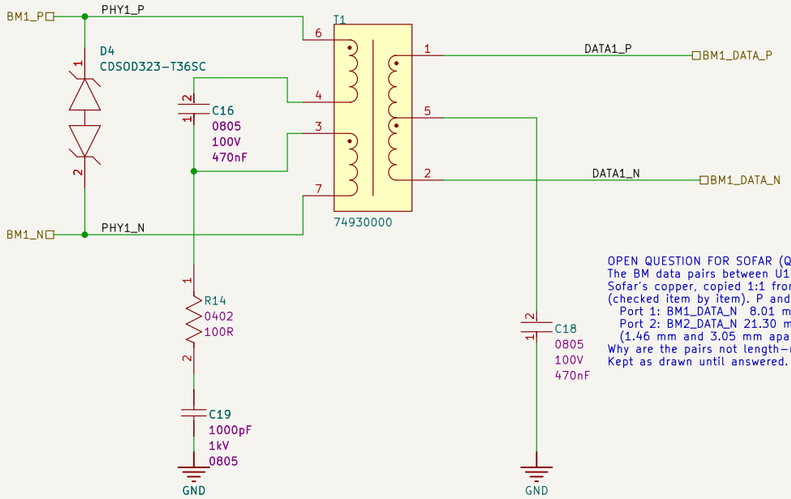
Sheet: /Top-Level Schematic/Load Switch/
File: Nereus_Pi_Shield_Adin2111_LoadSwitch.kicad_sch

Title: Load Switch

Size: A3	Date: 2026-11-11	**Rev: AA**
KiCad E.D.A. 10.0.7		Id: 4/7

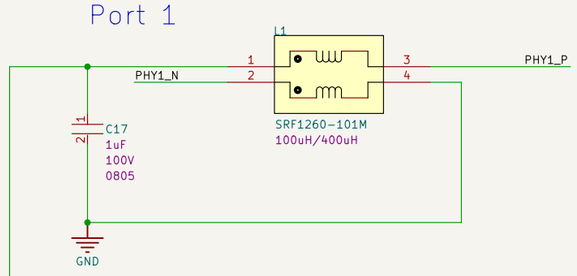
Power Over Dataline

Coupling Network – Port 1

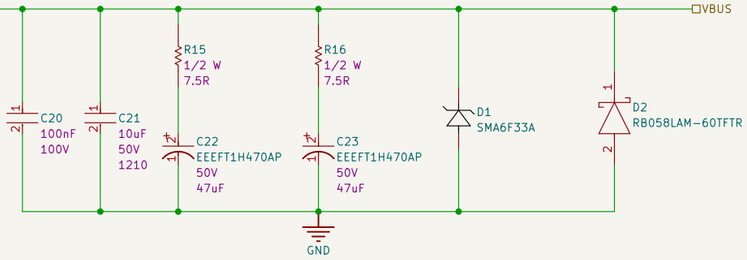


OPEN QUESTION FOR SOFAR (Q13, noted 2026-11-10)
The BM data pairs between U1 and T1 / T2 are
Sofar's copper, copied 1:1 from mote 000639-AB
(checked item by item). P and N differ in length:
  Port 1: BM1_DATA_N 8.01 mm, BM1_DATA_P 9.47 mm
  Port 2: BM2_DATA_N 21.30 mm, BM2_DATA_P 18.25 mm
  (1.46 mm and 3.05 mm apart)
Why are the pairs not length-matched?
Kept as drawn until answered.

Port 1



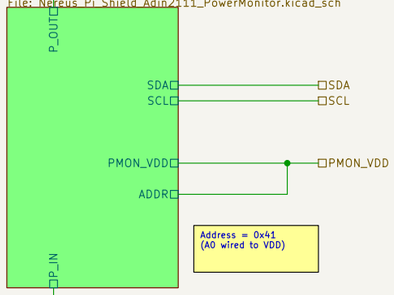
DC Connection + Circuit protections



Snubber resistors must be at least 1/2 W, to support hot-plugging.

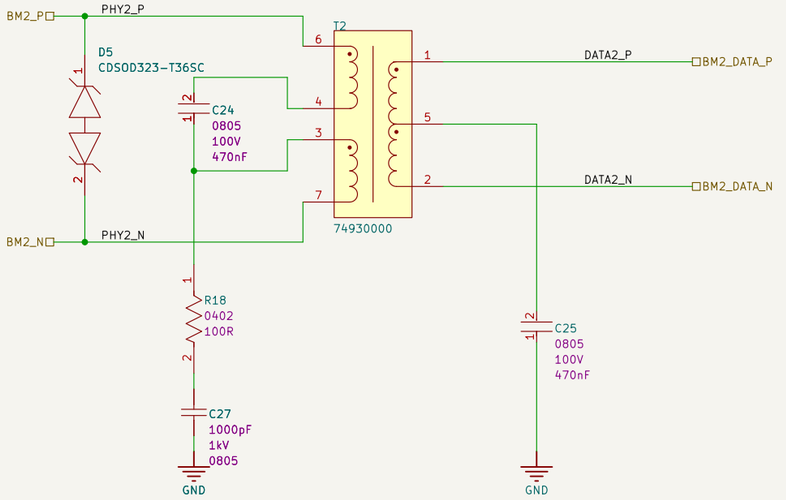
Power Monitor

File: Nereus_Pi_Shield_Adin2111_PowerMonitor.kicad_sch

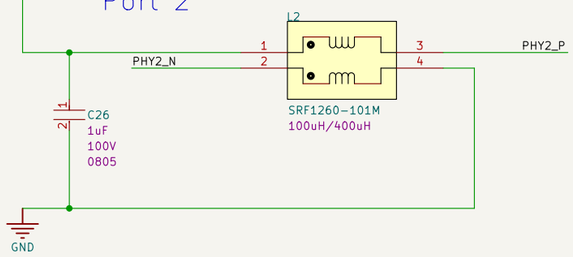


Address = 0x41
(A0 wired to VDD)

Coupling Network – Port 2



Port 2



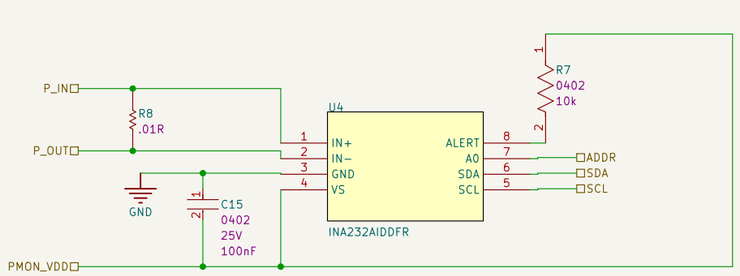
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
Nereus_Pi_Shield_Adin2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/
File: Nereus_Pi_Shield_Adin2111_PoDL.kicad_sch

Title: PoDL (Power over Data Line)

Size: A3	Date: 2026-11-11	**Rev: AA**
KiCad E.D.A. 10.0.7		Id: 5/7

I2C0 (Pi GPIO0/1) · Address 0x41 (A0 = VS)



Power Monitor

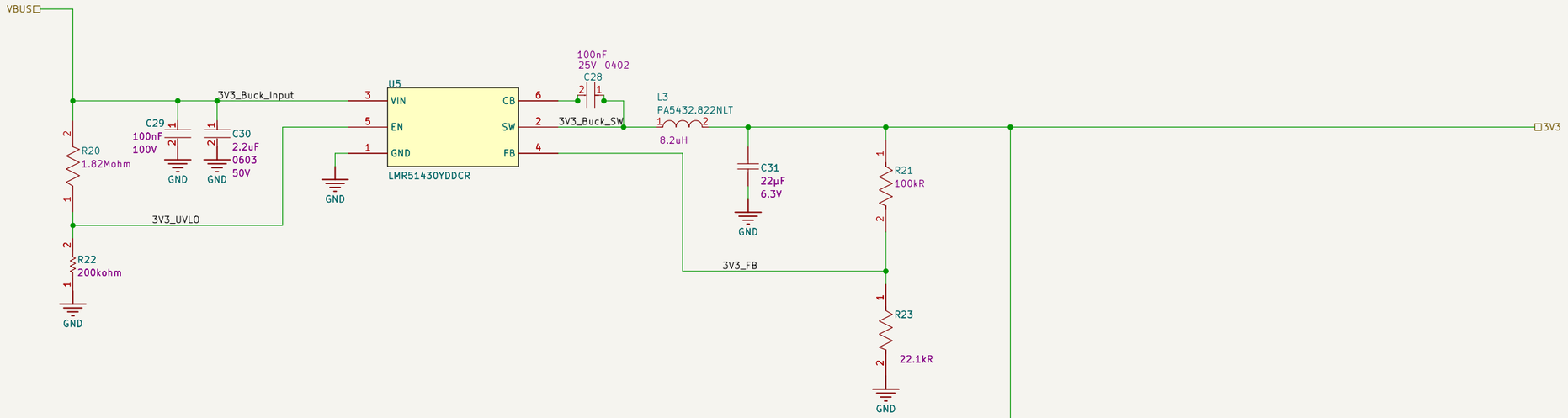
Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
Nereus_Pi_Shield_Adin2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/PoDL/Power Monitor/
File: Nereus_Pi_Shield_Adin2111_PowerMonitor.kicad_sch

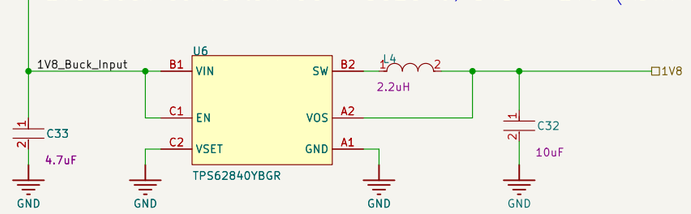
Title: Bus Power Monitor

Size: A3	Date: 2026-11-11	**Rev: AA**
KiCad E.D.A. 10.0.7		Id: 6/7

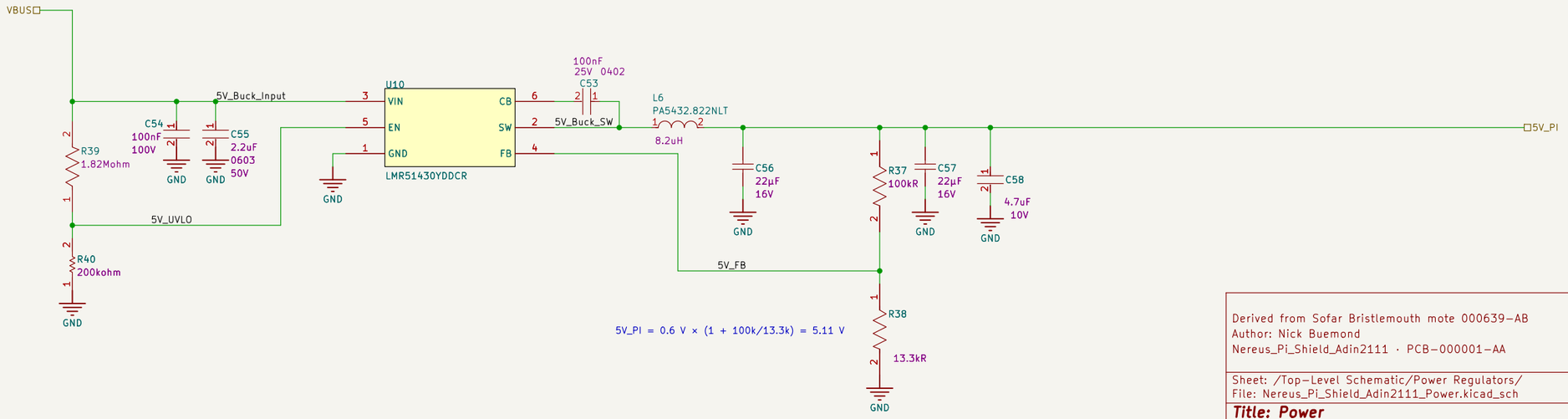
3V3 buck converter: U5 LMR51430, VBUS → 3V3



1V8 buck converter: U6 TPS62840, 3V3 → 1V8 (ADIN AVDD)



5V buck converter for the Pi: U10 LMR51430, VBUS → 5V_PI



Derived from Sofar Bristlemouth mote 000639-AB
Author: Nick Buemond
Nereus_Pi_Shield_Adin2111 · PCB-000001-AA

Sheet: /Top-Level Schematic/Power Regulators/
File: Nereus_Pi_Shield_Adin2111_Power.kicad_sch

Title: Power

Size: A3	Date: 2026-11-11	**Rev: AA**
KiCad E.D.A. 10.0.7		Id: 7/7